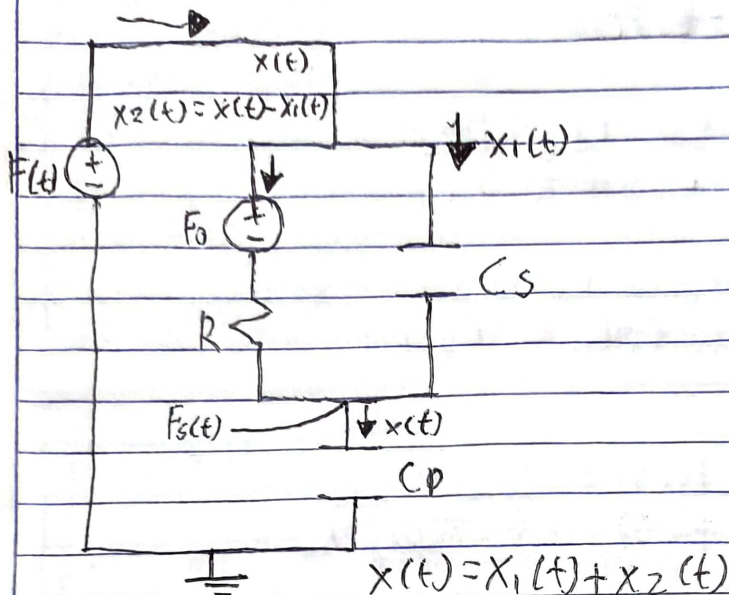
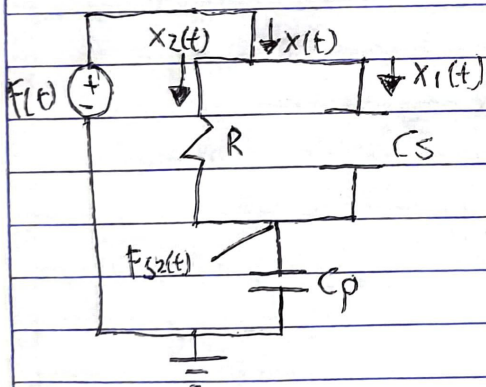


Circuito electrico



Función de transferencia

Análisis apagado F_0



$$x(t) = x_1(t) + x_2(t)$$

$$x_1(t) = C_s \frac{d[F(t) - F_s(t)]}{dt}$$

$$x(t) = C_p \frac{d[F_s(t)]}{dt}$$

$$x_2(t) = \frac{F(t) - F_s(t)}{R}$$

$$C_p \frac{dF_s(t)}{dt} = C_s \frac{d[F(t) - F_s(t)]}{dt} + \frac{F(t) - F_s(t)}{R}$$

$$C_p s F_s(s) = (C_p s [F_0 - F_s(s)] + F(s) - F_0(s))$$

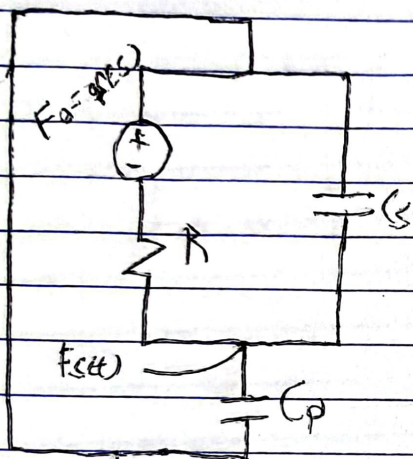
$$(C_p s + (C_p s + \frac{1}{R}) F_s(s) = (C_p s + \frac{1}{R}) F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{E}{C_p s + \frac{1}{R}}$$

$$\frac{C_p s + \frac{1}{R}}{C_p s + \frac{1}{R}}$$

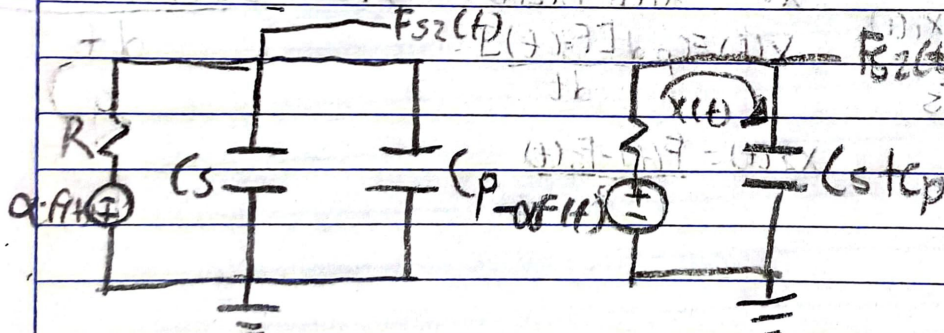
$$= \frac{C_p R s + 1}{C_p R s + 1}$$

$$\frac{F_s(s)}{F(s)} = \frac{C_p R s + 1}{C_p R s + 1}$$



$$\frac{F_s(s)}{F(s)} = \frac{C_p R s + 1}{R(C_p s + 1)}$$

$$F_s(s) = \frac{C_p R s + 1}{R(C_p s + 1)} F(s)$$



$$F(t) = R x(t) + \frac{1}{C_p} \int x(t) dt$$

$$F \propto F(s) = R x(s) + \frac{x(s)}{(Cst + p)s} \quad F_s(t) = \frac{x(s)}{(Cst + p)s}$$

$$F_s(s) = \frac{R(Cst + p)s + 1}{\alpha(Cst + p)s}$$

$$\frac{F_s(s)}{F(s)} = \frac{x(s)}{(Cst + p)s} = \frac{\alpha}{R(Cst + p)s + 1} (x(s))$$

$$F_s(s) = F(s) (s + F_s(s))$$

$$F_s(s) = \frac{(CsRs + 1)F(s) - \alpha F(s)}{R(Cst + p)s + 1}$$

$$\frac{F_s(s)}{F(s)} = \frac{CsRs + 1 - \alpha}{R(Cst + p)s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \left[1 - \frac{CsRs + 1 - \alpha}{R(Cst + p)s + 1} \right]$$

$$e(s) = 1 - 1 = 0$$

$$e(t) = 0V$$

$$e(s) = \lim_{s \rightarrow 0} s \cdot \frac{1}{s} \left[1 - \frac{CsRs + 1 - \alpha}{R(Cst + p)s + 1} \right] = 1 - \frac{1 - \alpha}{1}$$

$$= 1 - (1 - \alpha) = \alpha \quad e(t) = \alpha V = 0.25V$$

Estabilidad en lazo abierto

$$Z = R / (C + p) s + 1$$

$$Z = \frac{1}{(C + p)R} - 1 \Rightarrow Z_1 < 0$$

La respuesta es
asintóticamente estable